

SOLAR TILT OPTIMIZER

A forecast-informed web app for practical solar panel angle planning

What it does

Turns a location, panel area, and efficiency setting into a recommended tilt and azimuth.

Who it is for

Homeowners, students, and planners exploring solar orientation before engineering design.

Project overview

Solar Tilt Optimizer is a Streamlit website that estimates which panel tilt and orientation will capture the most forecasted solar energy at a chosen location. It combines location search, a seven-day weather and radiation forecast, and a constrained numerical optimization model. The output is designed to be easy to understand: key recommendations, expected energy, comparison charts, and a downloadable CSV.

How a visitor uses the app

- 1 Enter a city, postal code, or detailed address.
- 2 Set panel surface area and panel efficiency.
- 3 Choose a daily, quarterly, or fixed annual adjustment mode.
- 4 Review the recommended angles, forecasts, charts, and CSV export.

What the app shows

Recommendation	Optimal tilt in degrees and azimuth relative to due south.
Daily outlook	Hourly plane-of-array irradiance comparison between the optimized setting and a latitude-tilt baseline.
Weekly outlook	Seven-day predicted energy generation based on the forecast.
Energy map	A heatmap showing how expected energy changes across the allowed tilt and azimuth range.
Export	A CSV of hourly irradiance, estimated energy, and the recommended settings.
Quarterly mode	Four seasonal recommendations, with consecutive tilt changes limited to 30 degrees.

Data and calculation method

The website uses two free public services. OpenStreetMap Nominatim converts the visitor's location text into latitude and longitude. Open-Meteo provides hourly seven-day forecast values, including direct-normal irradiance, diffuse radiation, shortwave radiation, cloud cover, and temperature. Neither service requires an API key for this app.

Optimization model

For every hourly forecast point, the model estimates sunlight on the tilted panel using beam, diffuse-sky, and ground-reflected components. Solar altitude and bearing are calculated from the time and site. The optimizer selects the tilt and azimuth that maximize forecast plane-of-array irradiance after cloud and soiling adjustments.

The model uses the Liu-Jordan decomposition concept: total plane irradiance combines direct beam incidence, diffuse sky radiation on the tilted plane, and ground-reflected radiation. A low-tilt dust penalty is included as $L(\text{tilt}) = 0.12 \times \exp(-0.18 \times \text{tilt})$.

Variable	Allowed values	Reason
Tilt	10 to 60 degrees	Supports runoff/self-cleaning while respecting practical structural limits.
Azimuth	45 degrees east to 45 degrees west of south	Avoids extreme row-shading and retains a south-facing design focus.
Quarterly tilt step	Maximum 30 degrees	Keeps manual seasonal changes practical.

Reliability and limitations

When a live location or weather request is unavailable, the app remains usable with a clearly labelled Phoenix demo forecast. Recommendations are planning estimates, not a final engineering design. A qualified solar professional should verify site shading, roof strength, mounting hardware, electrical design, permits, utility rules, and manufacturer requirements before installation.

Run it or share it

The project can be run on one computer for testing, or published online for public use. A local address such as `http://localhost:8501` works only on the computer running the app. Publishing through Streamlit Community Cloud gives the project a public web address that can be sent to other people.

Run locally

1. Install Python. 2. Open PowerShell in the project folder. 3. Run `python -m pip install -r requirements.txt`. 4. Run `python -m streamlit run app.py`. The site opens at a local browser address, usually `http://localhost:8501`.

Publish a shareable website

- 1 Create a public GitHub repository.
- 2 Upload `app.py` and `requirements.txt` to the repository's main branch.
- 3 Sign in at Streamlit Community Cloud with GitHub and choose Deploy a public app from GitHub.
- 4 Set the branch to main and the main file path to `app.py`, then click Deploy.
- 5 Share the resulting `https://...streamlit.app` address. Visitors do not need Python.

Important note

Forecasts and third-party geocoding are live services. Their availability and output can change. The app labels its backup demo data whenever it cannot reach a live service. Do not use the tool as the sole basis for an installation decision.

Project files

File	Purpose
<code>app.py</code>	The complete Streamlit application, forecast integration, optimizer, charts, and CSV export.
<code>requirements.txt</code>	The Python packages Streamlit installs to run the app.
<code>README.md</code>	Short technical setup and model summary.

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